UNDER REVIEW

RESCIENCE C

Replication / Computational Neuroscience

[Re] An independent, provenance-carrying re-implementation of the PyRhO three-state channelrhodopsin-2 photocycle

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1 Introduction

Optogenetics drives or silences neurons by expressing light-gated ion channels such as channelrhodopsin-2 (ChR2) and illuminating them. Interpreting such experiments relies on models of the ChR2 photocycle, most commonly the three- and four-state Markov schemes introduced by Nikolic and colleagues[1], and on coupling the resulting photocurrent to a membrane model. PyRhO[2] is the reference open-source platform for this family of models: it fits and simulates three-, four- and six-state photocycles and couples them to neuron models through NEURON and Brian2.

This paper reports an *independent re-implementation* of PyRhO's three-state ChR2 photocycle, coupled to a single-compartment Hodgkin-Huxley membrane[3], and its validation against the reference. The re-implementation, `optosim`, is written from PyRhO's published equations and parameter table rather than by calling PyRhO, and is compared with PyRhO trace-against-trace.

The point of the exercise is not a new model — there is none here, and that is stated plainly throughout. It is twofold. First, to verify by independent re-implementation that the three-state equations as distributed in PyRhO can be transcribed correctly and reproduced to numerical precision. Second, to carry a single engineering discipline through the whole implementation: every numeric value the software emits is tagged with its origin — `MEASURED` (cited to a source), `DERIVED` (computed from cited quantities), `PREDICTED` (a model output) or `HEURISTIC` (a chosen constant with no source) — and that origin can only degrade through a computation, never be upgraded. A simulated series is `PREDICTED` by construction and there is no code path by which it can claim to be a measurement. This makes explicit, and machine-checkable, the distinction that reproducibility in computational neuroscience most often loses: which of the numbers driving a simulation anyone actually measured.

2 Methods

2.1 The model

The three-state photocycle has a closed state C, an open state O and a desensitised state D, with light-driven activation, light-independent decay and light-sensitive recovery:

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Code is available at https://github.com/fasn98/optosim – DOI 10.5281/zenodo.23177396..

$$\frac{dC}{dt} = G_r(\phi)\,D - G_a(\phi)\,C, \quad \frac{dO}{dt} = G_a(\phi)\,C - G_d\,O, \quad \frac{dD}{dt} = G_d\,O - G_r(\phi)\,D, \tag{1}$$

with $C + O + D = 1$ by construction, and the photon-flux-dependent rates

$$G_a(\phi) = k_a \frac{\phi^p}{\phi^p + \phi_m^p}, \qquad G_r(\phi) = G_{r0} + k_r \frac{\phi^q}{\phi^q + \phi_m^q}, \qquad G_d = \text{const.} \tag{2}$$

The photocurrent carries ChR2's inward rectification,

$$f_v(V) = v_1 \frac{1 - e^{-(V-E)/v_0}}{V - E}, \qquad I = g_0\,O\,f_v(V)\,(V - E), \tag{3}$$

with the removable singularity at $V = E$ handled by its analytic limit $f_v \to v_1/v_0$. All eleven ChR2 parameters ($g_0, \phi_m, k_a, k_r, p, q, G_d, G_{r0}, E, v_0, v_1$) are taken verbatim from PyRhO's fitted table (`modelFits['3']['ChR2']`) and cited to the file and line; the model form is attributed to Nikolic and colleagues[1]. The membrane is the classical Hodgkin-Huxley squid-axon model with its canonical parameters[3], and the photocurrent is added as an extra membrane current. A leaky integrate-and-fire neuron is implemented alongside the Hodgkin-Huxley one as a deliberately pessimistic baseline, the way a random data split is paired with a principled one, so that the difference between the two reports how much of any result comes from the biophysical detail rather than from the setup.

2.2 Coupling the absolute conductance to a per-area model

PyRhO reports g_0 as a whole-cell conductance (pS), whereas the Hodgkin-Huxley currents are per unit area (μA/cm^2). Bridging them needs a membrane area. Rather than choose one, the area is *derived* from a whole-cell capacitance C_{whole} and the specific membrane capacitance $C_m \approx 1\ \mu$F/cm^2 — a quantity whose near-constancy across cell types is itself a long-standing experimental result[4] — through area $= C_{\text{whole}}/C_m$. With a representative $C_{\text{whole}} = 100$ pF this gives 10^{-4} cm^2. The value is tagged `DERIVED`, and the report states that the photocurrent density scales linearly with $1/\text{area}$, so every spike count depends on it; the whole-cell capacitance is representative, not a measurement of any particular cell.

2.3 Provenance and the anti-fabrication guard

Parameters are `MEASURED` only with a source precise enough to find them again; uncited constants are `HEURISTIC` and listed explicitly on every run. In the three parameter sets, the Hodgkin-Huxley (8) and ChR2 (11) values are all cited; three of the six integrate-and-fire values (threshold, reset, refractory period) are uncited, because the integrate-and-fire model has no canonical parameters — and, consequently, any comparison of spike *counts* against it is a comparison against an arbitrary threshold, which the report says. A test walks the package's syntax tree and fails the build if any module that produces a scientific output can reach a random-number generator, so a fabricated or noise-perturbed value cannot enter a result unnoticed.

2.4 Validation design

Two questions are separated, because only one can be answered here. *Is the integration correct?* Under constant illumination the three rates are constant, so the system is linear and has an exact solution by matrix exponential; the fixed-step RK4 integrator is compared against it directly. *Does the implementation agree with PyRhO?* PyRhO's own integration of its own equations is the reference. PyRhO 0.9.4 does not import under

Python 3.12 (an `OrderedDict` incompatibility in a dependency); it was *not patched* — editing the reference until it agrees is not a validation — but run unmodified in an isolated Python 3.9 environment (numpy 1.19.5) to generate reference traces, which are archived with the code. A step of constant irradiance was applied at 1 and 10 mW/mm^2 for 200 ms, sampled at `dt = 0.01` ms (20001 samples).

3 Results

Numerics. The RK4 integrator reproduces the exact matrix-exponential solution to the limit of floating point: the maximum absolute error in the open fraction is 5.7×10^{-15} at 1 mW/mm^2 and 1.1×10^{-12} at 10 mW/mm^2, with state-conservation error of the same order. Integration error is therefore not a limit on anything the package reports.
Steady state. The converged open fraction matches PyRhO's documented closed-form steady state to a relative deviation of 1.1×10^{-12} (1 mW/mm^2) and 1.9×10^{-12} (10 mW/mm^2).
Transient, against PyRhO. The full time course agrees with PyRhO's own integration to a maximum absolute open-fraction deviation of 1.7×10^{-8} at 1 mW/mm^2 and 3.4×10^{-8} at 10 mW/mm^2 — four to five orders of magnitude below any precision to which the parameters are known, and well within the 10^{-3} agreement tolerance set in advance. The residual sits at the level of the reference's own `odeint` tolerance, so it bounds agreement rather than proving exact equality. This matters precisely because the steady state already agreed to 10^{-12}: a steady-state match constrains the rate matrix but does not uniquely pin each rate, so a compensating transcription error in two rates could pass it and surface only in the transient. Running PyRhO closed that gap.
Qualitative behaviour. The model reproduces ChR2's characteristic peak-then-plateau photocurrent, with a peak-to-plateau open-fraction ratio of 2.22 at 1 mW/mm^2 and 2.83 at 10 mW/mm^2, within the range reported for ChR2 — which is consistent with the known phenomenology and is *not* a measurement of agreement with any experiment.

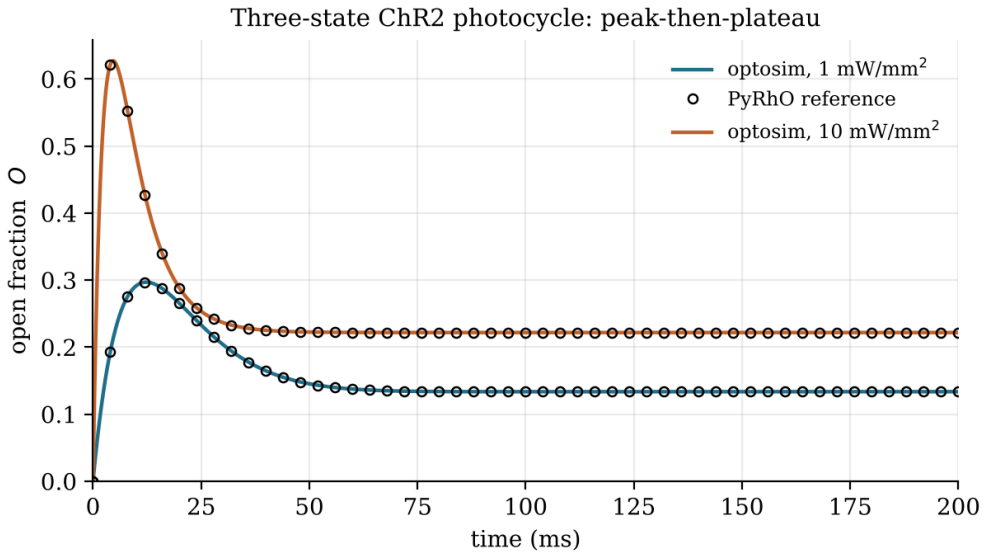


Figure 1. Three-state ChR2 open fraction under a step of constant irradiance at 1 and 10 mW/mm^2. `optosim` (lines) and PyRhO, run unmodified as the reference (circles), are indistinguishable at this scale; the maximum open-fraction deviation is 1.7×10^{-8} at 1 mW/mm^2 and 3.4×10^{-8} at 10 mW/mm^2.

Coupled model. On a 100 ms, 5 mW/mm^2 pulse the Hodgkin-Huxley neuron fires once and then settles at a depolarised plateau (depolarisation block), while the integrate-and-

fire baseline, lacking sodium inactivation, fires 35 times over the same pulse. The gap illustrates what the biophysical detail contributes; it does not say which model is correct, and the baseline's spike count rests on an uncited threshold, so the comparison is of mechanisms, not a calibrated prediction of either.

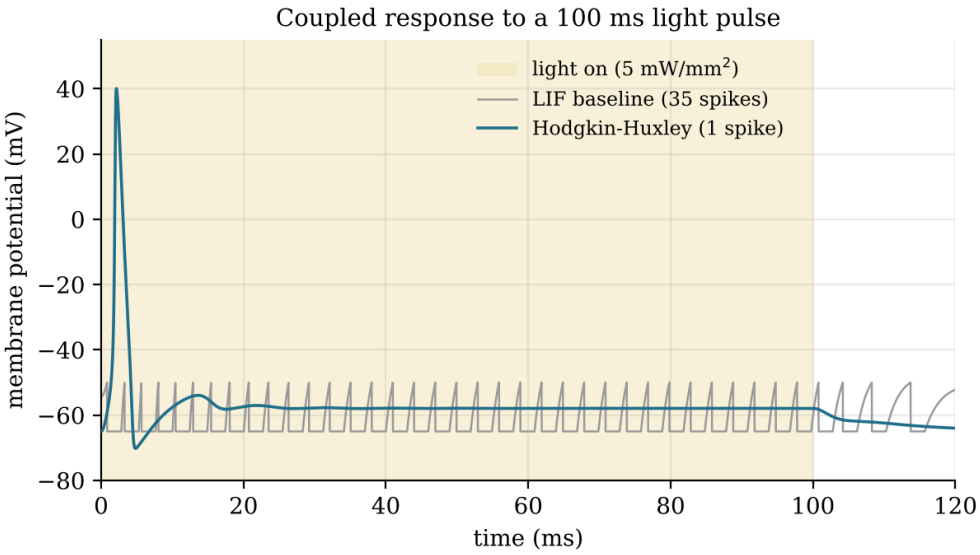


Figure 2. Membrane potential of the Hodgkin-Huxley neuron and the integrate-and-fire baseline under a 100 ms light pulse at 5 mW/mm^2 (shaded). The Hodgkin-Huxley neuron fires once and enters depolarisation block; the baseline, lacking sodium inactivation, fires 35 times. The baseline threshold is uncited, so the comparison is of mechanisms, not a calibrated prediction.

The full test suite (108 tests) passes, including the anti-fabrication guard and the exact-solution cross-check.

4 What this does and does not establish

Agreeing with PyRhO to 10^{-8} establishes that the three-state equations were *transcribed correctly* and can be reproduced independently to numerical precision. It does *not* establish that the three-state photocycle describes the real protein: that would require comparison against measured photocurrents, which has not been attempted. Two further limitations are stated rather than left to be inferred. The kinetics are Hodgkin-Huxley squid-axon kinetics at squid-axon temperature driving a mammalian opsin fit, with *no Q10 correction*, so the spike counts are not those the same protocol would produce at 37 °C. And irradiance is the value *at the opsin*: tissue absorption, scattering and depth attenuation are not modelled.

5 Conclusion

The PyRhO three-state ChR2 photocycle reproduces independently to numerical precision, confirming the equations as distributed. The contribution beyond that confirmation is methodological: an implementation in which every value carries a machine-checkable origin, a simulated series cannot be mislabelled as a measurement, and the uncited constants and unmodelled physics are reported on every run rather than buried. The code, the reference traces and the validation outputs are archived at DOI 10.5281/zenodo.23177396 and developed openly at https://github.com/fasn98/optosim.

6 AI assistance

This work was carried out with substantial assistance from Claude (Anthropic), used throughout the implementation, the design of the validation, the analysis and the drafting of this paper. The author set the goals, made the scientific and methodological decisions, directed and reviewed every computation reported here, and is responsible for the content, including any errors in it.

Claude is not an author. Authorship requires taking responsibility for the work, which a model cannot do; this follows ICMJE and COPE guidance.

The extent of that assistance is auditable rather than asserted: commits in the project repository carry `Co-Authored-By` trailers, so the development record can be inspected directly. Declaring it here applies to the authorship of the paper the same discipline the paper applies to its numbers — stating an origin rather than leaving it to be inferred.

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